

Conserved Quantities and Uncertainty as Phase-Space Area — A Unified Viewpoint of Symplectic Symmetry, Wick Rotation, and Stone’s Theorem

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Character of This Note

This is an observation paper. It does not propose a new physical theory.

This note does not modify standard quantum theory. It does not reject Wick rotation. It does not declare the Minkowski metric erroneous. It changes no observable prediction. Its entire claim is to **re-arrange and observe, from the single viewpoint of phase-space area**, structures that already exist within standard mathematics — the Heisenberg–Gabor uncertainty relation, Robertson’s inequality, the symplectic symmetry $\text{Sp}(2, \mathbb{R}) \cong \text{SU}(1, 1)$, Stone’s theorem, and Wick rotation.

We emphasize: what this note does is to confirm that the i appearing in Wick rotation and the i of the skew-adjoint generator in Stone’s theorem are described by **the same complex structure** of the standard complex Hilbert space, and to re-read this formal correspondence from the viewpoint of phase-space area (this is not a replacement). Consequently, the invariance of the speed of light and the causal (light-cone) order are **inherited** unchanged from standard theory. What this note adds to standard theory is not a new numerical prediction, but only a unified re-reading of known structures from one viewpoint.

This note does **not** claim any of the following:

- Modification of the mathematical predictions of standard quantum theory or special relativity
- Any assertion that Wick rotation or the Minkowski metric is “wrong”
- Derivation of the metric signature, the imaginary unit, or the complex structure
- Derivation of new physical constants, cross sections, or decay rates
- Proof of new mathematical theorems

Evaluation and interpretation are left to the reader.

Abstract

Classical signal theory (Fourier analysis, sampling) and quantum theory share a common currency when read as structures on the phase plane (q, p) : the action $\oint p dq$ (a conserved quantity) and the area spanned by uncertainty (fluctuation) are both measured as the symplectic area spanned by a conjugate pair. Taking this “area as common currency” as the starting point, this note observes three known facts from a single viewpoint.

First, the zero-point $\frac{1}{2}$ appearing in semiclassical quantization $\oint p dq = (n + \frac{1}{2})h$, the equality condition of Robertson’s inequality (the Gaussian ground state), the Maslov index (boundary character), and the lowest-weight representation of $SU(1, 1)$ (Bargmann index) appear as three facets of the common structure of half-integer weights of the metaplectic representation, the double cover of $Sp(2, \mathbb{R})$ (the three are not, however, numerically identical).

Second, the totality of area-preserving linear transformations forms $Sp(2, \mathbb{R}) \cong SU(1, 1)$; the symplectic area is kept invariant by the group action (on the representation-theoretic side, the Casimir corresponds as the representation label), and squeezing, chirp, and phase rotation are group actions. In particular, the non-compact subgroup (squeezing) is isomorphic to the Lorentz boost on phase space, with velocity appearing as $v/c = \tanh \eta$ and shape parameter $k = e^{2\eta}$.

Third, under the above symmetry (the stage of observation is throughout an even-dimensional phase space $\mathbb{R}^{2n}$ spanned by n conjugate pairs), the hyperbolic structure and the imaginary unit i appearing in Wick rotation and the Lorentz boost can be **re-read** in the language of the representation form that Stone’s theorem assigns to the generator of continuous evolution (skew-adjoint $= i \times$ self-adjoint). This reading is observationally equivalent to Wick rotation and to standard quantum theory (this note does not derive i or the metric signature). From this viewpoint, whether a given two-dimensional plane appears as a rotation (sign $+$) or a hyperbola (sign $-$) is relative to whether one treats the one-parameter subgroup acting on that plane as compact time evolution or as a non-compact evolution corresponding to analytic continuation/boost, in just the same way as the coordinate-dependence of the imaginary-time direction in Wick rotation. Also, because of the uncertainty floor $\frac{1}{2}$, there is no zero-area point-like state in phase space.

This note makes no new claim and is limited to a juxtaposed observation of these known structures from one viewpoint.

§1 Motivation: Area as Common Currency

1.1 The rectangular wave and the divergence of bandwidth

One of the most basic objects in signal theory is the rectangular pulse. The Fourier transform of a single rectangle $g(t) = A \text{rect}(t/\tau)$ of width τ and height A is

$$G(\nu) = A\tau \text{sinc}(\nu\tau), \quad \text{sinc}(x) \equiv \frac{\sin(\pi x)}{\pi x}$$

with a tail $|G(\nu)|^2 \sim 1/\nu^2$. Hence the mean-square bandwidth

$$\langle \nu^2 \rangle \propto \int \nu^2 |G(\nu)|^2 d\nu$$

diverges. The time–bandwidth rms product of the rectangle is infinite, and it **cannot attain** the minimum value (the $1/4\pi$ below) — it does not fall short of it but exceeds it without bound. As the price of perfectly sharp corners, the rectangle cannot maintain a stationary profile in a dispersive continuum. Incidentally, maximal energy concentration for finite time and finite band is given not by the rectangle but by the prolate spheroidal wave functions [21], and the simultaneous concentration of g and G is bounded by a Hardy-type limit [22].

To obtain a finite rms bandwidth, one needs smoothing of corners, band-limiting approximation, or an effective cutoff via discrete sampling. The sampling theorem (Shannon 1949) [7] states that a band-limited continuous signal can be reconstructed from discrete samples; for a sample interval a , the distinguishable frequencies are limited to $|\nu| \leq \nu_s/2$ (the Nyquist edge, $\lambda \geq 2a$ in wavelength). This note refers to discrete sampling only as one example of an effective cutoff.

1.2 Two areas: count and floor

The discussion of the rectangular pulse connects naturally to the theme of this note, “area.” When the stage on which an observer reads quantities is taken to be the phase plane of the conjugate pair (q, p) , the meaningful invariants are concentrated in the dimension of area (action):

- $\oint p dq$: the area enclosed by a closed orbit (action, a conserved quantity).
- the area of fluctuation (uncertainty): the symplectic area $\sqrt{\det \Sigma}$ of a pure Gaussian state (which equals $\Delta q \Delta p$ in a zero-correlation frame).

Both have the dimension of “(a quantity along q) $\times$ (a quantity along p)” (action = phase-space area). It is area — not length or volume — because q and p are a single conjugate pair, and what a symplectic transformation preserves is area.

However, the two share the same dimension but are not the same number. Since this distinction is used repeatedly here, we separate the three quantities explicitly.

Quantity	Value (in $\hbar$)	Meaning
Floor of the fluctuation area $\sqrt{\det \Sigma}$ ($= \Delta q \Delta p$ in zero-correlation frame)	$\hbar/2$	Lower bound satisfied by all states (Robertson)
Orbital action of the ground state $\oint p dq$	$\frac{1}{2}\hbar = \pi\hbar$	Semiclassical action of $n = 0$
Phase-space cell per state	$h = 2\pi\hbar$	Density of states $1/h$

For the harmonic-oscillator state n , the orbital action $\oint p dq = (n + \frac{1}{2})h$ grows with n , whereas the fluctuation area stays at $\hbar/2$ independent of n . The correct relation is

$$\oint p dq = n h + \frac{1}{2}h \quad (\text{count} \times \text{cell } h + \text{the leftover } \frac{1}{2}h)$$

and $\oint p dq$ is the total counting “how many minimum-uncertainty cells.” **When this note speaks of “area as common currency,” it means that both $\Delta q \Delta p$ and $\oint p dq$ are measured in the unit of the same symplectic area — not that their numerical values coincide.** The $\hbar/2$, $\pi\hbar$, $2\pi\hbar$ in the table above differ by factors 2π and 4π and must not be conflated. We adopt this distinction as a consistent viewpoint.

§2 Three Settings in Which the Zero-Point $\frac{1}{2}$ Appears

2.1 Semiclassical quantization

When momentum is quantized in a closed space (perimeter L), the phase area of one cycle is

$$\oint p dq = p_n L = \left(\hbar \cdot \frac{2\pi n}{L} \right) L = n h$$

so the conserved quantity is the number n of waves fitting in the closed space times h . In the semiclassical (WKB/EBK) approximation, a fractional part determined by the boundary character is added:

$$\oint p dq = \left(n + \frac{\mu}{4} \right) h$$

(μ is the Maslov index). For smooth confinement (harmonic), two turning points each contribute $\frac{1}{4}$, so $\mu = 2$, i.e. fractional part $\frac{1}{2}$. For a hard wall (infinite well), the idealization gives $\oint p dq = n h$ (no $+\frac{1}{2}$). But the infinite well is the non-physical idealization of an infinite potential, and real confinement is meaningful only as a limit of it. Note that even in the infinite well each state satisfies the Robertson floor (for the ground state $\Delta x \Delta p = \hbar \sqrt{(\pi^2 - 6)}/12 \approx 0.568\hbar > \hbar/2$). **That is, the uncertainty floor holds even in the infinite well, whereas the action fractional part $\frac{1}{2}$ does not appear there.** The two are independent quantities and must not be identified numerically. Yet for the harmonic oscillator the two are both connected to the metaplectic representation, and so appear in correspondence as a half-integer structure (§2.4).

2.2 Robertson's equality and the Gaussian ground state

Inserting $A = q$, $B = p$, $[q, p] = i\hbar$ into Robertson's inequality (Robertson 1929) [2]

$$\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

gives

$$\Delta q \cdot \Delta p \geq \frac{\hbar}{2} = \frac{h}{4\pi}.$$

This is the same theorem as the Gabor–Heisenberg inequality $\Delta t \cdot \Delta \nu \geq 1/4\pi$ of signal theory (Heisenberg 1927 [1], Gabor 1946 [4]). Writing the equality condition

$$(A - \langle A \rangle) |\psi\rangle = i\lambda (B - \langle B \rangle) |\psi\rangle \quad (\lambda \in \mathbb{R})$$

in q, p gives a first-order differential equation whose solutions are, up to translation, mean-momentum phase, width, and overall phase, restricted to the Gaussian form $\psi(q) \propto e^{-q^2/2\sigma^2}$. These are the minimum-uncertainty states attaining the floor $\hbar/2$; measuring the representative (the harmonic-oscillator ground state) by energy gives $E_0 = \frac{1}{2}\hbar\omega$, and in orbital action $\oint p dq = \frac{1}{2}h$.

Important remark (definition of “area” in this note): When a chirp (shear) $M_c = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix}$ is applied later (§3), a correlation $\text{Cov}(q, p) \neq 0$ is induced between q, p , and even for a pure Gaussian the simple marginal product

$$\Delta q \Delta p = \frac{\hbar}{2} \sqrt{1 + \beta^2} > \frac{\hbar}{2}$$

increases. That is, the marginal product $\Delta q \Delta p$ is not $\text{Sp}(2, \mathbb{R})$ -invariant, and saturates Robertson only in a zero-correlation frame (a tilted ellipse saturates

Schrödinger–Robertson but not Robertson). What is genuinely invariant under $\text{Sp}(2, \mathbb{R})$ is the square root of the determinant of the covariance matrix Σ , i.e. the **symplectic area** measured by the left side of the Schrödinger–Robertson form

$$(\Delta q)^2(\Delta p)^2 - \text{Cov}(q, p)^2 \geq \frac{\hbar^2}{4}$$

(equal to $\hbar/2$ for a pure Gaussian; the covariance-matrix formulation is Simon–Sudarshan–Mukunda [23]). This is the invariant captured by de Gosson’s quantum blob [20], and $\oint p dq$ is measured in this unit as well. **Hereafter, whenever this note says “area” it consistently means this symplectic area $\sqrt{\det \Sigma}$, while the frame-dependent marginal product $\Delta q \Delta p$ is distinguished as the value in a zero-correlation frame.**

2.3 $\text{SU}(1,1)$ lowest-weight representation

The Bargmann index characterizing the lowest-weight representation (the representation starting from the ground state) of the symmetry group $\text{SU}(1,1)$ discussed below is one of two values $k = \frac{1}{4}$ or $\frac{3}{4}$, from which the fractional part of $\oint p dq = (n + \frac{1}{2})\hbar$ is determined group-theoretically (Bargmann 1947 [9], Perelomov 1986 [10]).

2.4 Observation: the half-integer structure running through the three $\frac{1}{2}$ ’s

The $\frac{1}{2}$ -type values appearing in the Maslov index (boundary character), the Robertson floor (minimum-uncertainty state), and the $\text{SU}(1,1)$ lowest weight (representation theory) are **separate quantities arising from individually different theorems** — the value of the Maslov index itself depends on the system and boundary conditions (as in §2.1, in the infinite well the $\frac{1}{2}$ disappears), and the Robertson floor $\hbar/2 = \hbar/4\pi$ is a number of action dimension, differing even in dimension from the dimensionless weights $\frac{1}{4}, \frac{3}{4}$. Hence this note does not regard the three as “the same object.”

Yet that the three share a half-integer structure is not an unrelated coincidence: **all of them can be understood as structures connected to the metaplectic representation, the double cover, and the half-integer phase.** Namely, the metaplectic representation (Weil representation) [24] of the double cover $\text{Mp}(2, \mathbb{R})$ (metaplectic group) of $\text{Sp}(2, \mathbb{R})$ commonly underlies the three through its half-integer weight. The ground state (lowest-weight state) saturating the minimum symplectic area, the Maslov correction entering the semiclassical action, and the Bargmann index appearing in the $\text{SU}(1,1)$ lowest weight — all of these connect to the half-integer structure of this double cover.

Therefore this note neither over-identifies the three as “the same object arising from the same theorem” nor regards them as “three unrelated coincidences.” We read them as **a structure connected to the half-integer weight of the**

metaplectic representation. The §1.1 fact — that the rectangle (fractional part 0) cannot attain the floor $1/4\pi$ because of its rms bandwidth divergence, while the Gaussian (fractional part $\frac{1}{2}$) sits exactly at the floor — is also a facet of this structure. Semiclassical quantization $\oint p dq = (n + \frac{1}{2})h$ reads as “the count n (an integer, which mode) plus the unremovable leftover $\frac{1}{2}$ (the uncertainty).”

§3 The Area-Preserving Symmetry Group $\text{Sp}(2, \mathbb{R}) \cong \text{SU}(1, 1)$

3.1 Area-preserving linear transformations

Taking the phase-plane vector to be (q, p) and the symplectic form to be $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, an area-preserving linear transformation M satisfies

$$M^T J M = J \quad (\Longleftrightarrow \det M = 1 \text{ in } 2 \times 2).$$

In 2×2 the conditions $M^T J M = J$ and $\det M = 1$ are equivalent, and $\text{Sp}(2, \mathbb{R}) = \text{SL}(2, \mathbb{R})$. The following three operations are elements of this group:

- Squeeze: $M_s = \begin{pmatrix} e^{-r} & 0 \\ 0 & e^{+r} \end{pmatrix}$ (compress one axis, stretch the other; area-invariant)
- Rotation: $M_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ (phase rotation / time evolution)
- Chirp (shear): $M_c = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix}$ (add βq to p = instantaneous frequency $\omega_0 + \beta t$)

Representing the same group via the complex amplitudes $(a, a^\dagger)$, it appears as $\text{SU}(1, 1)$ ($a \rightarrow ua + va^\dagger$, $|u|^2 - |v|^2 = 1$).

3.2 Area invariance, representation label, group action

The generators of the Lie algebra $\mathfrak{su}(1, 1)$ are

$$\begin{aligned} K_0 &\propto a^\dagger a + \frac{1}{2}, & K_\pm &\propto a^{\dagger 2}, a^2, \\ [K_0, K_\pm] &= \pm K_\pm, & [K_+, K_-] &= -2K_0. \end{aligned}$$

K_0 (the seed of rotation / time evolution) contains the same $\frac{1}{2}$ as the zero-point of §2. The Casimir invariant $C = K_0^2 - \frac{1}{2}(K_+ K_- + K_- K_+)$ does not move under the group action. The role of the Casimir must, however, be stated with restriction. **The Casimir fixes the type of $\text{SU}(1, 1)$ representation and is kept invariant under the group action; it does not directly determine the excitation number n or the action integral itself.** In the single-mode, two-photon realization, the Bargmann index $k = \frac{1}{4}, \frac{3}{4}$ separates the even/odd

Fock sectors, while the excitation number n varies as a weight within the same representation. Hence this note treats the Casimir as a “representation-theoretic label of the area structure” and does not identify it with the action integral $\oint p dq$.

Under this restriction, the two-layer structure is stated as follows:

- **The Casimir C (group-invariant) = the labeling side that fixes the area structure.** It does not move under the group action.
- **The group action $M(r, \varphi, \theta) =$ the side of shape/allocation.** It stretches, tilts, and rotates the ellipse while preserving the symplectic area $\sqrt{\det \Sigma}$.

Defining the aspect ratio of the squeeze as $k \equiv \Delta \tilde{p} / \Delta \tilde{q} = e^{2r}$ with $r = \frac{1}{2} \ln k$, the area $\sqrt{\det \Sigma}$ (preserved independently of k) and the shape k (independent of area) are orthogonally decomposed as product and quotient.

3.3 Boost = non-compact subgroup = squeeze

$\text{Sp}(2, \mathbb{R})$ has one-parameter subgroups of distinct character. The rotation K_0 is compact (circle, trigonometric functions, returns) and reads as a resting internal clock. The squeeze $K_{\pm}$ system is non-compact (hyperbolic, cosh / sinh, does not return). The standard single-mode squeeze operator

$$S(\eta) = \exp\left(\frac{\eta}{2}(a^2 - a^{\dagger 2})\right)$$

(the generator $a^2 - a^{\dagger 2}$ is skew-Hermitian, so it is unitary as it stands) acts on the canonical variables $q = (a + a^{\dagger})/\sqrt{2}$, $p = (a - a^{\dagger})/(i\sqrt{2})$ (with $[q, p] = i$) as $S^{\dagger} q S = e^{-\eta} q$, $S^{\dagger} p S = e^{+\eta} p$, giving in the (q, p) basis, exactly and with normalization,

$$\begin{pmatrix} e^{-\eta} & 0 \\ 0 & e^{+\eta} \end{pmatrix}.$$

That is, the position-like axis is compressed by $e^{-\eta}$ and the momentum-like axis is stretched by $e^{+\eta}$. The hyperbolic structure $\cosh \eta, \sinh \eta$ is **isomorphic** to the 1+1-dimensional Lorentz transformation (the phase-space squeeze=boost correspondence is Han–Kim–Noz [25]), with

$$\frac{v}{c} = \tanh \eta = \frac{k-1}{k+1}, \quad k = e^{2\eta}.$$

At $k = 1$ ($\eta = 0$) it is at rest, and at $k \rightarrow \infty$ it reaches the speed of light. The shape parameter k can, formally, be put in correspondence with a relativistic velocity parameter.

Here we state two restrictions explicitly. First, **any non-compact one-parameter subgroup is isomorphic to the additive group** $(\mathbb{R}, +)$, so the identification $v/c = \tanh \eta$ is a choice of parametrization and does not by itself carry physical content. It remains an observation that boost and squeeze share the same non-compact structure (and the invariance of the Casimir under boost is a consequence of its being a representation label, not the invariance of the relativistic rest mass itself — the two are structurally similar but not the same).

Second, the squeeze as a **passive symplectic coordinate transformation** merely views the same physical state in different canonical coordinates and does not change the distribution of the excitation number. By contrast, acting with the **active squeeze operator** $S(r)$ on the vacuum produces a squeezed vacuum, which in the Fock basis is a superposition of even photon-number states — the photon-number distribution does change. When this note says “the area (Casimir / $\sqrt{\det \Sigma}$) is invariant even as the shape k is varied,” it refers to the invariance of the symplectic area, not that the active operation preserves the Fock-number distribution.

Fig. 1: The Gaussian ground state (blue, circle, $k = 1$, isotropic) and a squeezed state (red, ellipse, $k = e^{2\eta}$) in the phase plane (q, p) . The position axis is compressed by $e^{-\eta}$ and the momentum axis is stretched by $e^{+\eta}$, but the symplectic area $\sqrt{\det \Sigma} = \hbar/2$ is invariant under the $\text{Sp}(2, \mathbb{R})$ action. This squeeze is isomorphic to a Lorentz boost on phase space ($v/c = \tanh \eta$) (§3.3).

§4 A Generator-Theoretic Re-reading of the Imaginary Unit and the Hyperbolic Structure

4.1 Position of this section (preventing misreading)

This section does **not** reject Wick rotation or standard quantum theory. A real rotation $e^{\theta J}$ (eigenvalues $e^{\pm i\theta}$ since $J^2 = -I$; compact) and a hyperbolic evolution $e^{\eta K}$ (eigenvalues $e^{\pm \eta}$ since $K^2 = +I$; non-compact) are carried into each other by the substitution $\theta = i\eta$. This circle→hyperbola transition is the **standard form of Wick rotation** (Wick 1954 [12]). Hence all observable predictions of this section coincide with the standard Lorentzian/Euclidean framework. What we do is confirm the formal correspondence that Stone-type evolution and Wick-type analytic continuation are described by the same complex structure — not introduce a new i .

4.2 Where does the complex structure come from (making the premise explicit)

We make an important premise explicit. **Stone’s theorem** (Stone 1932 [8]; **the uniqueness of the representation of the canonical commutation relations is the Stone–von Neumann theorem** [26]) presupposes a

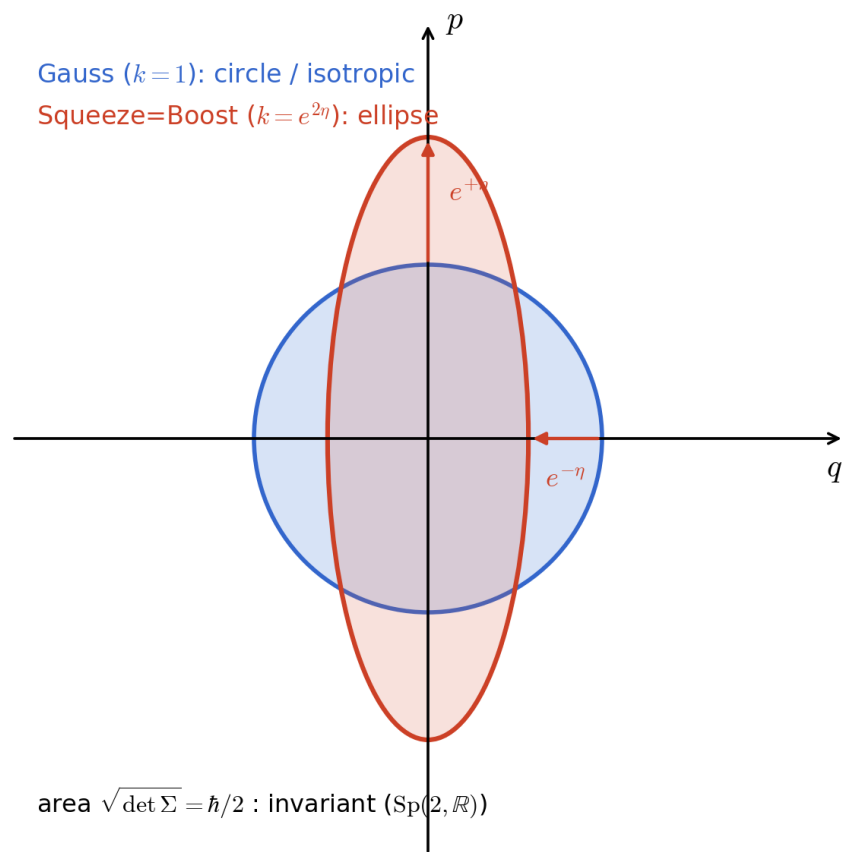


Figure 1: Fig. 1 Phase-space squeeze

strongly continuous one-parameter unitary group on a complex Hilbert space, and cannot be applied merely by projecting a real vector space. To speak of unitarity, the space must be equipped not merely as a real vector space but with a complex structure (an operator J satisfying $J^2 = -I$). Hence this note does not claim to derive the imaginary unit i . The i is **part of a presupposed structure** in the following sense.

The phase plane carries both the symplectic form ω (the $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ of §3, = area) and a positive-definite metric g . When there is a positive-definite metric compatible with the symplectic form, a **compatible almost complex structure** $\mathcal{J}$ satisfying

$$g(u, v) = \omega(u, \mathcal{J}v), \quad \mathcal{J}^2 = -I$$

is uniquely determined. This $\mathcal{J}$ is the imaginary unit i (the standard fact of Kähler structure / compatible triple, Cannas da Silva [27]). **The i is not derived from nothing out of real geometry; it is determined by the compatibility of two structures, the area ω and the metric g .** This is why a complex structure is unavoidably present in a space handling conserved probability, and it is the structure connecting the first half of this note (the symplectic phase plane) and the second half (the observational projection).

4.3 The form Stone's theorem assigns to the generator of evolution

A one-parameter evolution group that preserves observation (norm = probability preserving = unitary) can, under the complex structure of §4.2 and by Stone's theorem, take only the form

$$U(s) = \exp(sG), \quad G = \mathcal{J}H = iH \quad (H \text{ self-adjoint}).$$

That is, **the quantity generating continuous evolution is skew-adjoint (= $i \times$ self-adjoint), and in this sense bears i .**

We correct here a common misreading. The i in $p = -i\hbar\partial_q$ is not because “ p is an unobservable quantity” — momentum is an observable. This i is the manifestation of p being the generator of spatial translation. Likewise q generates translations in momentum space, and the two are symmetric. **Hence i is not “a mark intrinsic to an unobservable axis” but “the representation form of the generator of continuous evolution.”** Any self-adjoint quantity appears with i as the generator of the corresponding one-parameter group.

4.4 Re-reading the hyperbolic structure and the identification with Wick

Whether the “rotation” of a one-parameter group acting on a two-dimensional subspace is circular (cos / sin, compact, Euclidean, metric sign +) or hyperbolic

(cosh / sinh, non-compact, boost, metric sign $-$) is determined by the algebraic character of its generator — whether it is of $J^2 = -I$ type with eigenvalues $\pm i$, or of $K^2 = +I$ type with eigenvalues ± 1 . The two are carried into each other by the substitution $\theta = i\eta$, isomorphic to the Wick rotation of §4.1.

Therefore, whether a given two-dimensional plane appears as a circle (sign $+$) or a hyperbola (sign $-$) corresponds to **whether the observer treats the one-parameter subgroup acting on that plane as time evolution (compact rotation) or as analytic continuation/boost (non-compact)**. This note does **not derive** the metric signature or the Lorentzian structure here; it is limited to **re-reading** the hyperbolic structure already appearing in standard theory by associating it with the algebraic character of the generator. As a result:

- All observable predictions coincide with the standard Lorentzian/Euclidean framework.
- The invariance of the speed of light c and the causal (light-cone) order are **inherited** because they coincide with standard SR (reproduced by definition, not re-derived). This is a consequence of this note remaining within standard theory.

Same plane, two readings: compact (+) vs non-compact ($-$), by $\theta = i\eta$ (Wick)

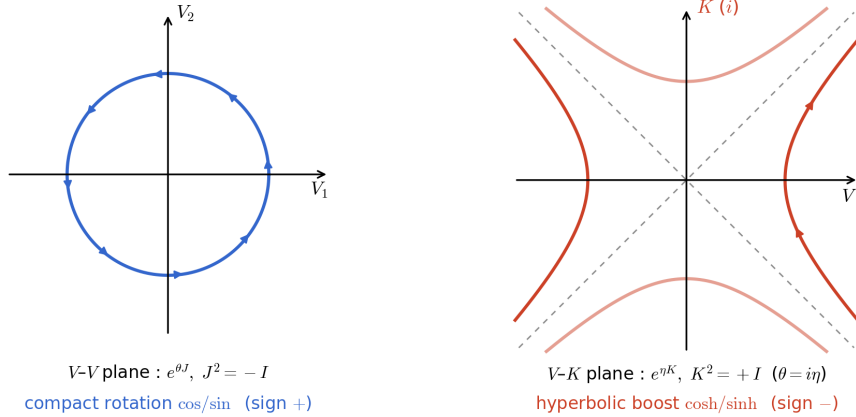


Figure 2: Fig. 2 Compact vs hyperbolic by projection

Fig. 2: One and the same two-dimensional plane splits into a circle (+) or a hyperbola ($-$) according to the algebraic character of the generator. Left: in a plane V - V within the observable subspace, the generator is of $J^2 = -I$ type, hence compact rotation (cos / sin, metric sign $+$). Right: in a plane V - K involving the kernel K (i -bearing evolution generator), it is of $K^2 = +I$ type, hence a hyperbolic boost (cosh / sinh, metric sign $-$). The two are carried into

each other by the substitution $\theta = i\eta$ (the standard form of Wick rotation) (§4.4).

4.5 Relativity of i

In which direction i appears — and hence which plane appears hyperbolic (sign $-$) — is relative to the observer’s choice of observable subspace. **This is just like how, in Wick rotation, which coordinate direction “imaginary time” points to depends on the choice of coordinates.** Taking the observable subspace in a different orientation moves the direction treated as the evolution generator. This relativity stays within the range observationally equivalent to standard theory and adds no new prediction. This note is limited to this observation.

§5 Extension to Multiple Axes, Composite Projection, and the Absence of Point-like States

5.1 Observational projection and the evolution side

In this section, the stage of projection is taken to be, as in §3–§4, an **even-dimensional phase space** $\mathbb{R}^{2n}$ (spanned by n conjugate pairs (q_i, p_i) and equipped with the symplectic form ω and the compatible almost complex structure $\mathcal{J} = i$ of §4.2). This is merely the natural extension of the phase plane (q, p) of §3 (a finite, boundaryless closed phase space is already used in $\oint p dq = nh$ of §2.1) and introduces no new mechanism. Since a conjugate momentum p_i is itself a generator of translation, the fact that a direction sent to the evolution side acts as a generator (§4.3) is intrinsic to the structure of this phase space.

In this space, since $\text{Sp}(2n, \mathbb{R})$ acts transitively on nonzero vectors, any axis can be chosen, **symmetrically**, as the observable subspace V or as its complement (the kernel K , the evolution side). One may send a single axis to the kernel, or send several axes together as a composite projection.

When a particular projection is fixed, the observable axes alone may fail to close a developmental/dynamical description. In that case, placing the kernel direction as an i -bearing evolution generator (by Stone’s theorem of §4, $i \times$ self-adjoint) makes the description consistent. In the description projected onto the observable space, this i -bearing generator structure can be read in correspondence with the hyperbolic structure of the plane containing that direction, or with the metric sign $-$ of standard theory. This note is limited to this observation.

To forestall misreading, we state what this note does **not** claim:

- It does not decide whether the causal structure is real or derivative.
- **It does not claim that i is introduced because causality would break down.** The order is the reverse: i is an adjustment to make

consistent a description that the observable axes alone cannot close, and the consistency of causality is only one aspect included in that consistency.

5.2 Order-independence of composite projection

Here we add a restriction. **General projection operators do not commute, and in the context of quantum measurement the order of non-commuting projections is physically important.** What this note calls “order-independence” is a claim restricted to the **idealization of sending mutually orthogonal independent subspaces to the evolution side (kernel) all at once.** Under this idealization, the direct-sum decomposition into observable/evolution sides is determined solely by the final specification of subspaces and does not depend on which orthogonal direction is sent first. At each step the observable dimension decreases by 1 and the evolution side increases by 1, but the sum

$$(\text{observable } k) + (\text{evolution side } N - k) = N$$

is preserved. The projection rule itself does not privilege a particular axis, but once a concrete observer has chosen V , the symmetry remaining in the description is restricted to the subgroup preserving V and K (of $\text{SO}(k) \times \text{SO}(N - k)$ type). That is, projection does not destroy the symmetry but contracts it to an apparently lower one.

5.3 On the all-evolution-side limit

Sending the observable dimension all the way to $k = 0$, all N axes become the evolution side (formally all imaginary). In this limit, no criterion for judging real or imaginary — “a mark relative to the observable directions” (§4.5) — remains, so for the observer there is no operation distinguishing all-imaginary from all-real. This is a limiting paraphrase of the relativity of §4.5 applied at $k = 0$, not a new claim.

Note that multiplying the whole state vector by a uniform phase is unobservable as a global phase, but this is a different operation from multiplying the coordinate axes / real structure by i , and this note does not identify the two.

5.4 Absence of zero-area point-like states (a consequence of the floor $\frac{1}{2}$)

First, to state it precisely: what the uncertainty relation forbids is a **zero-area quantum state** in phase space (a point-like state satisfying $\Delta q = \Delta p = 0$), not the origin as a mean value ($\langle q \rangle, \langle p \rangle = (0, 0)$). The latter is taken as an ordinary state. Hence the claim of this section is “there is **no zero-area point-like state** as an observable quantum state,” not “there is no coordinate origin.”

In this sense, $k = 0$ (all evolution side) cannot be placed as a “definite point-like state.” Phase space has a floor $\Delta q \cdot \Delta p \geq \hbar/2$ that forbids the state $\Delta q = \Delta p = 0$. Hence “all evolution side” and “total contraction” alike are meaningful only as **a limiting direction that cannot close to a single point because of the floor $\frac{1}{2}$** , not as a reachable point-like state.

This is consistent with the area invariance of §3. However much one direction is compressed by squeeze (boost), the symplectic area $\hbar/2$ remains and does not collapse to a point-like state. The degrees of freedom sent to the evolution side do not vanish but lie outside observation while keeping the floor.

§6 Relation to Existing Work, and What This Note Does Not Claim

6.1 Relation to existing work

Each component of this note is standardly established. The area interpretation of uncertainty is Robertson 1929 [2] / Schrödinger 1930 [3]; the time–band version is Gabor 1946 [4]. The phase-space quasi-probability distribution was discovered independently by Wigner 1932 [5] (quantum) and Ville 1948 [6] (signal). Sampling (Nyquist) is Shannon 1949 [7]. The relation between one-parameter unitary groups and generators is Stone 1932 [8]. The area-preserving transformation group and squeezed states are $\text{Sp}(2, \mathbb{R})$ / the metaplectic representation (Folland 1989 [17], Littlejohn 1986 [18]) and squeezed states (Stoler 1970 [19a], Yuen 1976 [19]). The representation-theoretic origin of $\frac{1}{2}$ is Bargmann 1947 [9] / Perelomov 1986 [10]. The boundary character is the Maslov index (Maslov 1972 / Arnold 1967 [11]). This note merely re-arranges these from the common viewpoint of area.

Regarding the interpretation of the imaginary unit i and the hyperbolic structure, we position this note’s stance against the following. **Toward each, this note asserts no conflict, and presents itself as an observationally equivalent re-reading.**

- **Wick rotation (Wick 1954 [12]) / Euclidean quantum gravity (Hartle–Hawking 1983 [13]):** This note reads the same $\theta = i\eta$ as a formal correspondence in which the i of the skew-adjoint generator of Stone-type evolution and the i of Wick-type analytic continuation are described by the same complex structure. The predictions coincide.
- **Relational quantum mechanics (Rovelli 1996 [14]) / QBism (Fuchs–Mermin–Schack 2014 [15]):** it shares the spirit of reading physical quantities as relations between observer and system.
- **Geometric algebra (Hestenes 1966 [16]):** it reads the i of quantum theory as a geometric object of spacetime. This note’s reading that “ i belongs to the generator structure” shares the spirit of detaching i from being an intrinsic attribute of coordinates, but the mechanism (the requirement

of unitarity by Stone's theorem) differs.

6.2 What this note does not claim

- Modification of the mathematical predictions of standard quantum theory or special relativity (this note is observationally equivalent to them).
- Any assertion that Wick rotation or the Minkowski metric is “wrong” (this is a reinterpretation, not a denial).
- **Derivation of the imaginary unit i , the complex structure, or the metric signature** (as in §4.2, i is a presupposed compatible almost complex structure determined by the compatibility of the area ω and the metric g).
- Any mention or proof of where the arrow of time comes from (this note does not do so, leaving it to the interpretation of existing standard theory).
- Derivation of new physical constants, scattering cross sections, decay rates, or new particles.
- Proof of new mathematical theorems.

6.3 What this note records

- That the conserved quantity $\oint p dq$ and the uncertainty $\sqrt{\det \Sigma}$ are measured in the unit of the same symplectic area of the phase plane (though the numbers differ: §1.2), and that the half-integer structure appears in the three settings of the Robertson floor, the Maslov index, and the $SU(1,1)$ lowest weight, not as the same object but as three facets of the metaplectic representation.
- That in the area-preserving symmetry group $Sp(2, \mathbb{R}) \cong SU(1,1)$, the Casimir is the representation-theoretic label of the area structure, the group action carries shape/allocation, and boost = non-compact subgroup = squeeze ($v/c = \tanh \eta$, $k = e^{2\eta}$).
- That under the compatible almost complex structure $\mathcal{J} = i$ determined by the compatibility of the area ω and the metric g , Stone's theorem assigns the form $i \times$ self-adjoint to the generator of continuous evolution. By this, the hyperbolic structure of Wick rotation / Lorentz boost can be **re-read in the language of the observational description**, and this reading is observationally equivalent to standard quantum theory (not a derivation). Whether a given plane appears as a rotation (+) or a hyperbola (−) is relative to the observer's projection.
- That, because of the uncertainty floor, there is no zero-area point-like state.

For the avoidance of doubt: **what this note means by “identification” is the formal correspondence that Stone-type evolution and Wick-type analytic continuation can be described using the same complex structure of a complex Hilbert space; it does not newly derive Wick rotation, the Lorentzian metric, or the complex structure.**

6.4 Open problems (stated explicitly as unresolved)

The following are not treated here, or are left to a separate paper. We state them explicitly for honesty.

1. **The mechanism determining the codimension of the observable subspace:** which codimension an observer's projection selects (including the grounds for spacetime appearing as $3 + 1$ in standard theory) is not explained within this framework (§5.1).
2. **Deepening the origin of the complex structure:** in §4.2, i was taken to be the compatible almost complex structure determined by the compatibility of the area ω and the metric g . To argue from more basic premises why this compatibility (compatible triple) is unavoidable for observation.
3. **Low-dimensional toy models and explicit forms:** to write out explicitly, for $N = 2, 3$, the $V \oplus K$ decomposition, the effective inner product, and the almost complex structure, and to display the form of Stone's generator $G = iH$.
4. **Visualization:** the phase-plane ellipses (Gaussian vs squeezed state) and the boost correspondence (Fig. 1), and the schematic of the circle/hyperbola distinction on the V - V / V - K planes (Fig. 2) are included. A more detailed illustration of the multi-axis projection is left to future work.

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